

VINCENT WREN

Houston, TX 77025

📞 779-245-4631 ✉ wrenvin@gmail.com 🌐 [vincent-wren](https://vincent-wren.com) 🌐 vincentwren.com

Education

University of Houston

May 2027

Honors BS of Computer Science — Minors in Architecture, Data & Society

GPA: 3.7

- **Relevant Coursework:** History of the Built Environment, Data Structures, Algorithms, Operating Systems, Space & Movement in the City (In Progress)

Vrije Universiteit Amsterdam

Jan. 2025 – Jun. 2025

Computer Science Exchange Student

- **Relevant Coursework:** Software Development, Machine Learning

Technical Skills

GIS & Spatial Analysis: ArcGIS Pro, ArcGIS CityEngine, QGIS, ArcGIS Online, Field Maps, GeoAI, Kepler.gl

Data & Visualization: Network Analysis, Georeferencing, OSM, GTFS, 3D Visualization

Programming: Python, JavaScript, Java, C++, Git/GitHub

Languages: English (Native), Dutch (Intermediate)

Projects

Midtown Houston Complete Streets Visualization | *ArcGIS CityEngine*

Fall 2025

- Procedurally modeled a 3D retrofit of the Louisiana & Smith corridor to visualize future-state concepts.
- Redesigned digital Right-of-Way (ROW) to transform 5-lane arterials into multimodal, transit-rich corridors.
- Demonstrated spatial feasibility of Road Diet interventions to prioritize safety using existing street widths.

Houston Transit Equity & Frequency Study | *ArcGIS Pro*

Fall 2025

- Mapped the METRO high-frequency transit network using GTFS data to identify critical service corridors.
- Computed accessibility metrics by overlaying 0.25-mile walk buffers onto Block Group population data.
- Visualized service coverage gaps to identify socioeconomic disparities in access to frequent transit.

Downtown Houston Land Use Analysis | *ArcGIS Pro*

Fall 2025

- Quantified CBD built environment to compare vehicle vs. pedestrian land allocation in the urban core.
- Synthesized municipal and OSM data to categorize road, parking, sidewalk, and park surface areas.
- Determined 71.9% of downtown land is for vehicles vs. 9.2% for pedestrians, highlighting policy needs.

Campus Bike Parking Utilization Field Survey | *ArcGIS Online*

Fall 2025

- Surveyed 74 bike rack locations via ArcGIS Field Maps to collect attribute data on capacity and usage.
- Identified critical capacity failures where library racks exceeded 100% usage while dorms remained empty.

Houston Amenity Equity Analysis | *QGIS*

Summer 2025

- Performed spatial joins in QGIS to correlate amenity density with Census racial demographic data.
- Revealed that majority-Black neighborhoods house 7.9% of the population but only 1.3% of grocery stores.

Experience

Lazarus Energy

Jul. 2025 – Present

Project Invent Manager

Houston, TX

- Lead 36 students through a year-long human-centered design invention program.
- Mentor teams through the full engineering design process, from research to pitch.

Gehl

Oct. 2024

Field Data Collector

Houston, TX

- Collected observational data for the Downtown Houston Public Life Survey.
- Documented public space usage patterns to support the 2025 downtown redesign.

Lazarus Energy

Oct. 2023 – Jan. 2025

IT Intern

Houston, TX

- Developed a real-time data management web app to enhance scheduling efficiency.
- Automated employee hour tracking using the Google Sheets API.

City of Houston

Jun. 2024 – Aug. 2024

TechConnect Intern

Houston, TX

- Facilitated STEM activities for youth to promote digital equity in under-resourced areas.
- Collaborated with professionals at TechConnect Fairs; earned Apple Teacher certifications.

Leadership

No Higher No Wider I-10 | *Website Manager*

Feb. 2024 - Present

- Maintain an advocacy website opposing I-10 freeway expansion.

BikeHouston | *Data Team Dashboard Developer*

Jan. 2024 – Aug. 2024

- Assisted in developing a data visualization dashboard for cycling infrastructure in Houston.